

done inclass T-Critical Value Using t-distribution table

Case I

$n=15$, Confidence interval = 90%, left-tailed

Significance level = 1 - Confidence interval

$$= 1 - 0.90$$

$$= 0.10$$

degrees of freedom = $n-1$

$$= 15-1$$

$$= 14$$

Important Steps to find out t-critical value

Step 1 ÷ Select the row for one-tailed

Step 2 ÷ Select the column for significance level (here it is 0.10)

Step 3 ÷ Then select the row for degrees of freedom (here it is 14)

Obtained value at the intersection point of all three steps is your t-critical value

done
inclass

Case II

$n=25$, Confidence interval = 95%, right-tailed

Significance level = 1 - Confidence level

$$= 1 - 0.95$$

$$= 0.05$$

degrees of freedom = $n-1$

$$= 25-1$$

$$= 24$$

Case - III $n = 35$, Confidence Interval = 99%, Two-tailed test

Significance level = $1 - \text{Confidence interval}$

$$= (1 - 0.99) = 0.01$$

$$= 0.005$$

degrees of freedom = $n - 1$

$$= 35 - 1$$

$$= 34$$

T-Critical value = 2.72

Numericals (done in class)

1) A factory has a machine that dispenses 80 ml of fluid in a bottle. An employee believes that average amount of fluid is not 80 ml. Using 40 samples, he measures average amount dispensed by the machine to be 78 ml with standard deviation of 2.5

(a) State the null and alternate hypotheses

(b) At a 95% confidence interval level, is there enough evidence to support the idea that the machine is not ^{working} properly? ~~working~~

Soln:

Given

1) Population mean $\mu = 80 \text{ mL}$

Sample mean ($\bar{x}$) = 78 mL

Sample size (n) = 40

Sample standard deviation (s) = 2.5 mL

Significance level (α) = 0.05

You have Sample size of 40, which is relatively large (usually $n \geq 30$ is considered)

Standard deviation (σ) is

problem gives Sample Standard deviation ($s = 2.5 \text{ mL}$) but don't have population Standard deviation (σ)

① State null Hypothesis

• Null Hypothesis (H_0): $\mu = 80 \text{ mL}$ (The average amount is 80 mL)

• Alternate Hypothesis (H_1): $\mu \neq 80 \text{ mL}$ (The average amount is not 80 mL)

2. Calculate the Test Statistics (t-value).

Using the formula

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Substitute the values:

$$t = \frac{78 - 80}{\frac{2.5}{\sqrt{40}}}$$

First, Calculate the denominator:

$$\frac{s}{\sqrt{n}} = \frac{2.5}{\sqrt{40}} = 0.395$$

Now, calculate t-value

$$t = \frac{-2}{0.395}$$

3. Find the Critical Value

• Since it is two-tailed test at $\alpha = 0.05$, the critical values for 39 degrees of freedom ($df = n - 1$) can be found using t-distribution table

• Let's find t-critical values for 39 degrees of freedom at 95% Confidence interval ($\alpha/2 = 0.025$ on each tail) is approximately ± 2.022

4. Decision Rule

• If absolute value of the Computed t-value is greater than the critical t-value ($|t| > 2.022$), reject null hypothesis

Since σ (population standard deviation) is not given, we would use t-test instead of z-test even though the sample size is large

T-distribution accounts for the additional uncertainty due to using sample

standard deviation (s) rather than population standard deviation (σ)

T-test is more appropriate test here because we don't know the population standard deviation

5. Conclusion

The Computed t -value is -5.06

Since $|t| = 5.06 > 2.022$, We reject the null hypothesis H_0 .

Final Answer

There is enough evidence at 95% Confidence level to conclude that the machine is not dispensing an average of 80ml. The machine is likely not working correctly.

2 tail vs 1 tail test

$\mu \neq 50$

Not equal

Non-directional

When alternate hypothesis looks like this i.e

$H_0: \mu = 50$ vs $H_1: \mu \neq 50$

that means

μ can be less than 50

or μ can be more than 50

it is non-directional

LTT

$H_1: \mu < 50$

'Smaller'

'lower'

'reduced'

'under'

'below'

'less than'

RTT

$H_1: \mu > 50$

'Higher'

'more'

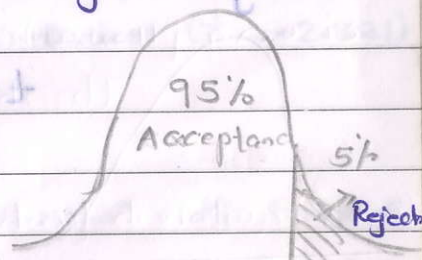
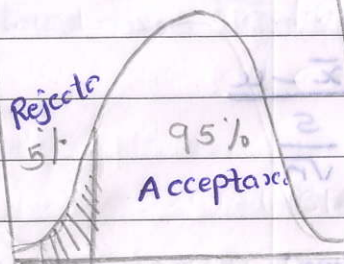
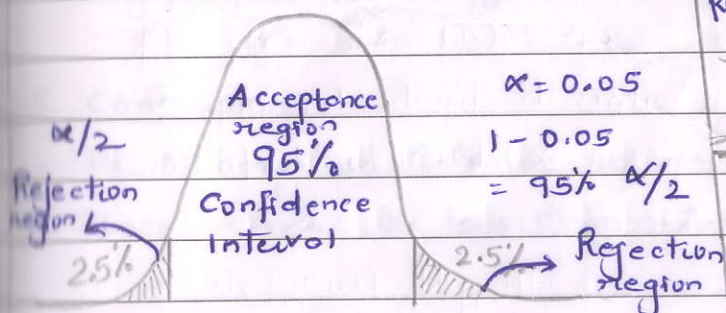
'exceed'

'increased'

'above'

'larger'

'greater than'



If t -critical value lies within this range then we fail to

2. A Company manufactures car batteries with an average life span of 2 or more years. An engineer believes that this value to be less. Using 10 samples, he measures the average life span to be 1.8 years with standard deviation of 0.15.

(a) State the null and alternate hypothesis

(b) At a 99% Confidence level, is there enough evidence to discard the null hypothesis?

Solⁿ:

- population mean (μ) = 2 years
- Sample mean ($\bar{x}$) = 1.8 years
- Sample Size (n) = 10
- Standard deviation (s) = 0.15 years
- Significance level (α) = 0.01

1. State the Hypothesis

- NULL Hypothesis (H_0): $\mu = 2$ years (The average lifespan is 2 years)
- Alternate Hypothesis (H_1): $\mu < 2$ years (The average lifespan is less than 2 years)

2. Calculate the Test Statistics (t-Value)

Using the formula

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Substitute the values:

$$t = \frac{1.8 - 2}{\frac{0.15}{\sqrt{10}}}$$

First Calculate the denominator

$$s = 0.15 = \frac{0.15}{\sqrt{10}} = 0.0474$$

Now, Calculate the t-value

$$t = \frac{-0.2}{0.0474}$$

$$= -4.22$$

3. Find the Critical Value:

Since it is one-tailed t-test at $\alpha = 0.01$ with 9 degrees of freedom ($df = n - 1$)

t-distribution table typically gives the positive critical value as +2.821

However, since this is left-tailed test (We're testing if the sample mean is less than the population mean), the critical value we need to consider is -2.821

Note: * If you're finding +2.821 in the table, that's correct for a right-tailed test

* For a left-tailed test we should use -2.821 as the critical value

4. Decision Rule

If the computed t-value is less than ^{the} critical value ($t < -2.821$) reject the null hypothesis

5. Conclusion:

- The Computed t-value is -4.22
- Since -4.22 is less than -2.821, we reject the null hypothesis H_0

Final answer

There is enough evidence at the 99% Confidence level to support the engineer's claim that the average lifespan of the batteries is less than 2 years.

3) Generalized Steps for a T-Test

1. State the hypothesis

- NULL hypothesis (H_0): The parameter (e.g. mean) is equal to specific value.

- Alternate hypothesis (H_1)

- For one-tailed Test:

- Left-Tail $H_1: \mu < \mu_0$

- Right-Tail $H_1: \mu > \mu_0$

- For a two-tailed Test:

$$H_1: \mu \neq \mu_0$$

2. Calculate the Test Statistics

- Use the t-test formula to calculate test statistics

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

where $\bar{x}$ is the Sample mean

μ_0 is the hypothesized population mean

s is the Sample Standard deviation

n is the sample size

3. Find the Critical values

Determine the Critical t-value from the t-distribution table based on significance level (α) and degrees of freedom ($df = n - 1$).

- For one-tailed test:

- Use α to find the critical value for the specific tail (left or right)

- For a two-tailed test:

4. Decision Rule

One-tailed Test

- Left tail: Reject H_0 if the calculated t-value is less than the negative critical t-value.

- Right tail: Reject H_0 if the calculated t-value is greater than the positive critical t-value.

Two-tailed Test

- Reject H_0 if the absolute value of the calculated t-value is greater than the critical t-value.

(absolute value of a number^(x) is denoted by $|x|$)

5. Conclusion

Based on evidence decision rules, conclude whether there is sufficient evidence to reject the NULL hypothesis H_0 in the favour of alternate hypothesis.

Why absolute value is used in two-tailed test?

In a two-tailed test, you are checking for deviations from hypothesized mean in both the direction (either higher or lower). The critical region is on both ends of the distribution.

Therefore you compare the absolute value of t-value to the critical value t-value to see if it falls into either tail.

- If $|t| > \text{Critical t-value}$, you reject the null hypothesis H_0 .
- If $|t| \leq \text{Critical t-value}$, you fail to reject H_0 .

Using absolute value ensure that you're considering the magnitude of the difference from the hypothesized mean, irrespective of the direction (positive or negative)

Example

Suppose you calculate a t-value of -2.5 for your test and critical t-value is 2.009 for two-tailed Test at a 0.05 significance level

- The absolute value of the calculated t-value is $|-2.5| = 2.5$
- You would compare 2.5 with 2.009
- Since $2.5 > 2.009$ you reject the null hypothesis H_0

If you didn't take absolute value and just compared the negative t-value to the positive critical value, it wouldn't make sense in context of a two-tailed test, where deviations in either direction are significant

note in class

How to find p-value from t-Test Statistics

① Fred was testing null hypothesis with mean $= 0$

$H_0: \mu = 0$ Versus $H_a: \mu \neq 0$. His samples had 8 observations and his test statistics was $t = 2.98$. Assuming the condition for inference were met. What is the approximate p-value for Fred's test

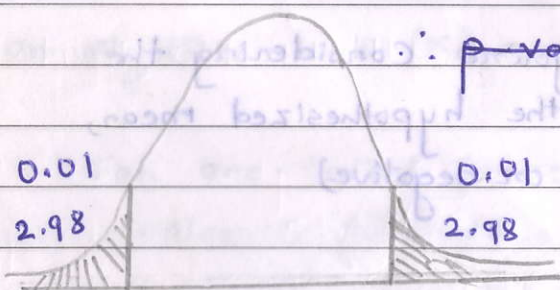
$$n = 8 \quad df = n - 1$$

$$t = 2.98 \quad = 8 - 1$$

$$df = 7$$

In the row of 7 search for the value which is close to t-critical value i.e here $t = 2.98$

numb



2) Caterina was testing $H_0: \mu = 0$ versus $H_a: \mu \neq 0$ with sample of 6 observations. Her test statistics $t = 2.75$. Assume that the conditions for inferences were met.

What is the approximate p-value for Caterina's test

$$df = n - 1$$

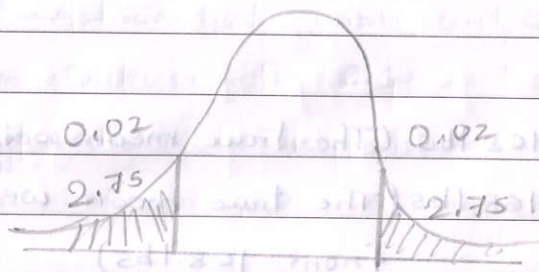
$$= 6 - 1$$

$$df = 5$$

$t = 2.75$ check in the 5th row of t-distribution

Critical value table

we get $p = 0.02$ where value is matching



$$\therefore p\text{-value} = 0.04$$

3) The average weight of all residents in town XYZ is 168 lbs. A nutritionist believes the true mean to be different. She measured the weight of 36 individuals and found the mean to be 169.5 lbs with standard deviation of 3.9.

(a) State null hypothesis and alternate hypothesis

(b) At 95% Confidence interval, is there enough evidence to discard null hypothesis? (Use p-value method)

Soln: population mean (μ) = 168 lbs

Sample mean ($\bar{x}$) = 169.5 lbs

Sample size (n) = 36

Sample standard deviation (s) = 3.9

Significance level (α) = 0.05 (Two-tailed test)

Step 1: State the hypothesis

Null hypothesis (H_0): $\mu = 168$ lbs (The true mean weight is 168 lbs)

Alternate hypothesis (H_1): $\mu \neq 168$ lbs (The true mean weight is different from 168 lbs)

Step 2

Calculate the test Statistics

The test Statistics (t) is calculated using the formula

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$t = \frac{169.5 - 168}{\frac{3.9}{\sqrt{36}}}$$

First, Calculate the denominator

$$= \frac{3.9}{\sqrt{36}}$$

$$= \frac{3.9}{6}$$

Now, calculate the t-value

$$t = \frac{1.5}{0.65}$$

$$= 2.31$$

Step 3: Find the critical value

For a two-tailed test $\alpha = 0.05$, with degrees of freedom

$$df = n - 1$$

$$= 36 - 1$$

Using t-distribution table, critical value for 35 df = 35 at $\alpha/2 = 0.025$ (each tail)

The critical value for a two-tailed test is approximately ± 2.030

Step 4: Find p-value

Using the t-distribution table or calculator for $df = 35$

• The t-value calculated is 2.31

• The corresponding p-value for $t = 2.31$ is approximately 0.027
two a two-tailed test (Using table i got answer as 0.025)

Step 5: Decision Rule

- If p-value is $p\text{-value} \leq \alpha (0.05)$, reject null hypothesis H_0
- otherwise do not reject H_0

Step 6: Conclusion

• The Calculated p-value is 0.027

• Since $0.027 < 0.05$, we reject null hypothesis H_0

Final answer

There is a sufficient evidence at a 0.05 significant level to conclude that true mean of weight of the residents is different from 168 lbs

* ~~How to Calculate p-value for a one-tailed test using z table~~

Under

Z-test

- 1) The mean expenditure per customer at a fire store is \$85.00 with a standard deviation of \$9.00. If the random sample of 40 customers is taken, what is the probability that the sample average expenditure per customer for this sample will be \$87.00 or more?

Solution Figure out whether we are looking for one-tailed test or two-tailed test

One-tailed test

When you are testing for the possibility of relationship in one direction and completely disregarding the possibility of a relationship in the other direction

Example

Testing whether the sample mean is greater than or less than certain value

Two-tailed test

When you are testing for the possibility of relationship in ^{both} one directions

Example: Testing whether sample mean μ is different (either greater or less) than certain value

Here we are asked to find the probability that the sample average expenditure per customer for this sample will be \$87.00 or more

We are only interested in testing whether the sample mean is greater than \$87.00. This indicates we are concerned with one direction (greater than \$87.00), which means this is one-tailed test

Step 1: Identify the given information

Population mean (μ): \$ 85.00

Population standard deviation (σ): \$ 9.00

Sample size (n): 40

Sample mean ($\bar{x}$): \$ 87.00

Step 2: Calculate the standard error of the mean (SEM)

Standard error of the mean is calculated using the formula

$$SEM = \frac{\sigma}{\sqrt{n}}$$

$$= \frac{9.00}{\sqrt{40}}$$

$$= \frac{9.00}{6.324}$$

$$= 1.422$$

Step 3: Calculate Z-score

Z-score is calculated using the formula

$$Z = \frac{\bar{x} - \mu}{SEM}$$

$$= \frac{87.00 - 85.00}{1.422}$$

$$= 1.407$$

$$\approx 1.41$$

Step 4: Find the probability corresponding to the Z-score

Cumulative probability for $Z = 1.41$ is approximately 0.9207

Step 5: Calculate the probability of getting a sample mean of \$ 87.00 or more

Since Z table gives cumulative probability that the sample mean

$$p(\bar{x} \geq 87.00) = 1 - 0.9207 \\ = 0.0793$$

probability that the Sample average expenditure ~~is~~ per customer for this Sample will be \$87.00 or more is approximately 0.0793 or 7.93%

Dec 2023 (done in class)

Gmks

2) On an examination given to the students at a large no. of different schools, the mean grade was 74.5 and standard deviation 8.0. At one particular school, where 200 students took examination, the mean grade was 75.9. Discuss the significance of the result at the 0.05 level from view point of

Given Data

(i) One tailed test

(ii) Two tailed test

• population mean (μ) = 74.5

• population standard deviation (σ) = 8.0

• Sample mean ($\bar{x}$) = 75.9

• Sample size (n) = 200

• Significance level (α) = 0.05

Solⁿ:

Step 1: State the hypothesis

(i) one-tailed Test

• Null Hypothesis (H_0): $\mu \leq 74.5$ (The mean grade at the school is less than or equal to population mean)

• Alternate Hypothesis (H_1): $\mu > 74.5$ (The mean grade at the school is greater than the population mean)

(ii) Two-tailed test

• Null Hypothesis (H_0): $\mu = 74.5$ (The mean grade at the school is equal to the population mean)

• Alternate Hypothesis (H_1): $\mu \neq 74.5$ (The mean grade at the school is different from population mean)

Step 2: Calculate the Test Statistics

Since the population standard deviation is known, we use the z-test. The Test Statistics z is calculated using the formula:

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$= \frac{75.9 - 74.5}{\frac{8.0}{\sqrt{200}}}$$

First, calculate the Standard error:

$$\text{Standard error} = \frac{8.0}{\sqrt{200}}$$

$$= \frac{8.0}{14.14}$$

$$= 0.566$$

Now Calculate z-value

$$z = \frac{1.4}{0.566}$$

$$= \frac{1.4}{0.566} = 2.47$$

Step 3: Find Critical value and p-value

(i) one-tailed test:

- For one-tailed test at $\alpha = 0.05$, the critical value is approximately 1.645

Significance level $\alpha = 0.05$ means that there is 5% in the right tail of the distribution

Cumulative probability for rest of the distribution (to the left)

$$= 1 - \alpha$$

$$= 1 - 0.05$$

$$= 0.95$$

* Look up the z-score for 0.95 (or closet value) in the z-table:
In in z-table you look up the cumulative probability of 0.9500 (or closet value to 0.95)

The z-score corresponding to 0.9500 is 1.645

Here is the small section of the z-table for the cumulative probabilities near 0.95

Z-score	Cumulative probability
1.64	0.9495
1.645	0.95
1.65	0.9505

$$\frac{0.9495 + 0.9505}{2}$$

$$= 0.95$$

$$\frac{1.64 + 1.65}{2}$$

$$= \frac{3.29}{2}$$

$$= 1.645$$

(ii) p-value for $Z=2.47$ is 0.0068

as per the table we get

$$0.9932$$

$$1 - 0.9932 = 0.0068$$

(ii) Two-tailed test

For two tailed test $\alpha = 0.05$

In a two tailed test we are splitting the significance level α into two halves, one for left tail and one for right tail

If $\alpha = 0.05$ then $\frac{\alpha}{2} = 0.025$ in its tail

* Find z-score for upper tail

• we need to find the z-score corresponding to cumulative

- Use the z-table to find the z-score where the cumulative probability is close to 0.975

* Look up the z-score for ~~0.975~~ ^{0.975} in the z-table
 In the z-table look up the cumulative probability of 0.975. The z-score corresponding to ~~0.975~~ 0.975 is approximately 1.96

Here is the small section of z-table for cumulative probabilities near 0.975

Z-score	Cumulative probability
1.95	0.9744
1.96	0.975
1.97	0.9756

$$\frac{0.9744 + 0.9756}{2} = 0.975$$

$$\frac{1.95 + 1.97}{2} = \frac{3.92}{2} = 1.96$$

- p-value for $Z = 2.47$ in a two-tailed test is approximately 0.013

Step 4

(i) One-tailed test

If $Z > 1.645$, reject the null hypothesis H_0

Since $Z = 2.47 > 1.645$ and $p\text{-value} = 0.0068 < 0.05$, we reject null hypothesis

(ii) Two-tailed test

If $|Z| > 1.96$, reject the null hypothesis

Since $Z = 2.47 > 1.96$ and $p\text{-value} = 0.0136 < 0.05$, we reject null hypothesis

Step 5.

(i) One-tailed test

There is sufficient evidence at 0.05 Significance level to conclude that the mean grade at school is significantly higher than the population mean of 74.5

(ii) Two-tailed test

There is sufficient evidence at 0.05 Significance level to conclude that the mean grade at a school is significantly different from population mean of 74.5

Z-score	Cumulative Probability
1.96	0.9750
1.9	0.9744
1.95	0.9755
1.96	0.9758

Value for $Z = 1.96$ in two-tailed test is approximately 0.025

One-tailed test
If $Z > 1.96$, reject the null hypothesis
Since $Z = 2.1$ and $p\text{-value} = 0.018$, we reject the null hypothesis

Two-tailed test
If $|Z| > 1.96$, reject the null hypothesis
Since $Z = 2.1$ and $p\text{-value} = 0.018 < 0.05$, we reject the null hypothesis

Questions on two Sample t-test

June 2023 (done in class)

1. Find the value of t-test for following set of values (4 marks)
~~7, 2, 9, 8~~ 7, 2, 9, 8 and 1, 2, 3, 4

Soln: To perform a t-test, we need to decide what kind of t-test to use. Since you have provided two sets of values we can perform independent sample t-test

Let's go step-by-step to find the t-test

Data

Group 1: 7 2 9 8

Group 2: 1 2 3 4

Step 1: State the null hypothesis

- NULL hypothesis (H_0): There is no significant difference between the means of the two groups i.e. $\mu_1 = \mu_2$
- Alternate hypothesis (H_1): There is a significant difference between the means of two groups i.e. $\mu_1 \neq \mu_2$

Step 2: Calculate the mean of each group

Group 1: mean of Group 1

$$\bar{x}_1 = \frac{7+2+9+8}{4} = \frac{26}{4} = 6.5$$

Group 2: mean of Group 2

$$\bar{x}_2 = \frac{1+2+3+4}{4} = \frac{10}{4} = 2.5$$

Step 3: Calculate the variance of each group.

Variance formula

$$\text{Variance} = \frac{\sum (x - \bar{x})^2}{n-1}$$

Group 1 Variance

$$= \frac{(7-6.5)^2 + (2-6.5)^2 + (9-6.5)^2 + (8-6.5)^2}{4-1}$$

$$= \frac{(0.5)^2 + (-4.5)^2 + (2.5)^2 + (1.5)^2}{3}$$

$$= \frac{0.25 + 20.25 + 6.25 + 2.25}{3}$$

$$= \frac{29}{3}$$

$$= 9.67$$

Group 2 Variance

$$= \frac{(1-2.5)^2 + (2-2.5)^2 + (3-2.5)^2 + (4-2.5)^2}{4-1}$$

$$= \frac{(-1.5)^2 + (-0.5)^2 + (0.5)^2 + (1.5)^2}{3}$$

$$= \frac{2.25 + 0.25 + 0.25 + 2.25}{3}$$

$$= \frac{5}{3}$$

$$= 1.66$$

Step 4: Calculate the pooled Standard deviation

The pooled standard deviation S_p is used because the sample sizes are equal

$$S_p = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}$$

$$= \sqrt{\frac{(4-1)(9.67) + (4-1)(1.67)}{4+4-2}}$$

$$= \sqrt{\frac{3(9.67) + 3(1.67)}{6}}$$

$$= \sqrt{\frac{29.01 + 5.01}{6}}$$

$$= \sqrt{\frac{34.02}{6}}$$

$$= \sqrt{5.67}$$

$$= 2.38$$

Step 5: Calculate the t-statistics

the formula for t-statistics is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$= \frac{6.5 - 2.5}{2.38 \sqrt{\frac{1}{4} + \frac{1}{4}}}$$

$$= \frac{4.0}{2.38 \sqrt{\frac{1}{4} + \frac{1}{4}}}$$

$$= \frac{4.0}{2.38 \sqrt{\frac{1}{2}}}$$

$$= \frac{4.0}{2.38 \times 0.707} = \frac{4.0}{1.68} = 2.38$$

Step 6: Degrees of freedom and Critical value

degrees of freedom (df) for an independent samples t-test is:

$$df = n_1 + n_2 - 2$$

$$= 4 + 4 - 2$$

$$= 6$$

Using a t-distribution table for 6 degrees of freedom and significance level $\alpha = 0.05$ for a two-tailed test, critical value is approximately ± 2.447 .

Step 7: Decision

- If $|t| > \text{Critical value}$, reject H_0 .

Here $t = 2.38$ and critical value is 2.447 .

Since $|t| = 2.38$, we fail to reject the null hypothesis.

Step 8: Conclusion

There is not enough evidence to suggest a significant difference between the means of two groups at 0.05 significance level.

June 2023

2) Internal test marks of the students are as follows (7 marks)

10.5, 9, 7, 12, 8.5, 6.5, 8, 11, 9.9

Mean population score is 12 and standard deviation is 1.80.

Is the mean value for Student significantly different from the mean population value? Let alpha value is 0.05. So corresponding t-value is 2.62.

Q1: Given Data

- Internal test marks of students: 10.5, 9, 7, 12, 8.5, 6.5, 8, 11, 9.9
- Mean population score (μ) = 12
- Population standard deviation (σ) = 1.80
- Significance level (α) = 0.05
- Critical t-value = 2.262

We need to check if the mean value of students significantly differ from ^{mean} population mean value using a t-test

Step 1: State the hypothesis

- NULL hypothesis (H_0): The mean of the student's score is equal to the population mean, $\mu = 12$
- Alternative hypothesis (H_1): The mean of the student's score is different from population mean, $\mu \neq 12$

Step 2: Calculate the Sample mean

We need to calculate the sample mean ($\bar{x}$) of the given student's marks

$$\bar{x} = \frac{\sum x_i}{n} = \frac{10.5 + 9 + 9.7 + 12 + 8.5 + 6.5 + 8 + 11 + 9.9}{9}$$

$$\bar{x} = \frac{85.1}{9}$$

$$\bar{x} = 9.46$$

So, the sample mean is approximately 9.46

Step 3: Calculate Sample Standard deviation

Let's calculate sample standard deviation (s) using the formula

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

Standard deviation

First, calculate $(x_i - \bar{x})^2$ for each value {when you are working with population standard deviation then use n }

$$\bullet (10.5 - 9.46)^2 = 1.1025$$

$$\bullet (9 - 9.46)^2 = 0.2116$$

$$\bullet (9.7 - 9.46)^2 = 0.0576$$

$$\bullet (12 - 9.46)^2 = 6.4516$$

- $(8 - 9.46)^2 = 2.1316$

- $(11 - 9.46)^2 = 2.3716$

- $(9.9 - 9.46)^2 = 0.1936$

Now sum them up

$$\begin{aligned}\sum (x_i - \bar{x})^2 &= 1.1025 + 0.2116 + 0.0576 + 6.4516 + 0.9216 + 8.7561 \\ &\quad + 2.1316 + 2.3716 + 0.1936 \\ &= 22.1978\end{aligned}$$

Now divide by $n-1$

$$= 9 - 1 = 8$$

$$S^2 = \frac{22.1978}{8}$$

$$S^2 = 2.7747$$

$$S = \sqrt{2.7747}$$

$$S = 1.665$$

Step 4: Calculate the Test statistics (t-value)

The formula for test-statistics is

$$t = \frac{\bar{x} - \mu}{\frac{S}{\sqrt{n}}}$$

Where

$$\bar{x} = 9.46 \text{ (Sample mean)}$$

$$\mu = 12 \text{ (population mean)}$$

$$S = 1.665 \text{ (Sample standard deviation)}$$

$$n = 9 \text{ (Sample size)}$$

Substitute the values

$$t = \frac{9.46 - 12}{\frac{1.665}{\sqrt{9}}}$$

Step 5: Compare t-value with Critical value

The calculated t-value is -4.58 and critical t-value is ± 2.262 at a significance level of 0.05 for a two-tailed test.

Step 6: Decision Rule

If $|t| > 2.262$, reject the null hypothesis

$|t| < 2.262$, fail to reject the null hypothesis

Since $|t| = 4.58$ is greater than 2.262 , we reject null hypothesis

Step 7: Conclusion

There is sufficient evidence to conclude that the mean score of students is significantly different from population mean at the 0.05 level of significance level.

June 2023

3) A sample size of 16 people is taken. Suppose a claim is made that the average number of days a person spends a vacation is more than equal to 5 days based on sample of 16 people whose mean come out to be 9 days. Sample standard deviation is found to be 3 days at a level of significance of 0.05 and t-value for right-tailed test comes out to be 1.753 . Calculate the value of t and check null hypothesis

Solⁿ:

Given Data

- Sample size $n = 16$
- Sample mean $(\bar{x}) = 9$ days
- Hypothesized mean $(\mu) = 5$ days
- Sample standard deviation $(s) = 3$ days
- Significance level $(\alpha) = 0.05$
- Critical t-value for right-tailed test = 1.753

Step 1: State the hypothesis

- Null hypothesis (H_0): The average number of days a person spends on vacation is $\mu = 5$ days

Alternate hypothesis (H_1): The average number of days a person spends on vacation is greater than 5 days ($\mu > 5$ days)

This is one-tailed right test

Step 2: Calculate test-statistics value (t-value)

The formula for t-statistics is

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

where

- $\bar{x} = 9$ (Sample mean)
- $\mu = 5$ (population mean)
- $s = 3$ (Sample standard deviation)
- $n = 16$ (sample size)

$$t = \frac{9 - 5}{\frac{3}{\sqrt{16}}}$$

$$= \frac{4}{\frac{3}{4}} = \frac{4}{0.75} = 5.33$$

$$= 1.753$$

$$= 1.75$$

Step 3: Compare the t-value with Critical value

- The Calculated t-value is 5.33
- The Critical t-value is 1.753 for a one-tailed test

Step 4: Decision Rule

- If $t > 1.753$, reject the null hypothesis H_0
- If $t \leq 1.753$, fail to reject the null hypothesis H_0

Since $t = 5.33$ is greater than 1.753 , we reject the null hypothesis

Step 5: Conclusion

There is sufficient evidence at the 0.05 significance level to conclude that average number of days a person spends on vacation is significantly greater than 5 days

June 2023

4) The marks ~~for~~ of boys and girls are as follows (6 marks)

Boys: 12, 14, 10, 8, 16, 5, 3, 9, 11

girls: 21, 18, 14, 20, 11, 19, 8, 12, 13, 15

Is there any significant difference between the marks of girls and boys? Assume $\alpha = 0.05$ t -value (critical) = 2.11

Soln:

Step 1: State the hypothesis

- Null hypothesis (H_0): There is no significant difference between the marks of boys and girls ($\mu_1 = \mu_2$)
- Alternate hypothesis (H_1): There is significant difference between marks of boys and girls ($\mu_1 \neq \mu_2$)

This is two-tailed test

Step 2: Calculate the means of both groups

Mean of boys marks ($\bar{x}_1$)

$$\bar{x}_1 = \frac{12+14+10+8+16+5+3+9+11}{9}$$

$$= \frac{88}{9}$$

$$\approx 9.78$$

Mean of girl's marks $\bar{x}_2$

$$\bar{x}_2 = \frac{21+18+14+20+11+19+8+12+13+15}{10}$$

$$= \frac{151}{10}$$

$$\approx 15.1$$

Step 3: Calculate standard deviation of both groups

Standard deviation of boy's group

$$S_1 = \sqrt{\frac{\sum (x_i - \bar{x}_1)^2}{n_1 - 1}}$$

$$S_1 = \sqrt{\frac{(12-9.78)^2 + (14-9.78)^2 + (10-9.78)^2 + (16-9.78)^2 + (5-9.78)^2 + (3-9.78)^2 + (9-9.78)^2 + (11-9.78)^2}{8}}$$

$$S_1 = 4.12$$

Standard deviation of girl's group

$$S_2 = \sqrt{\frac{\sum (x_i - \bar{x}_2)^2}{n_2 - 1}}$$

Step 4: Calculate t-statistics

The formula for the t-test is

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

where

$$\bar{X}_1 = 9.78$$

$$\bar{X}_2 = 15.1$$

$$S_1 = 4.12$$

$$S_2 = 4.28$$

$$n_1 = 9 \text{ (number of boys)}$$

$$n_2 = 10 \text{ (number of girls)}$$

$$t = \frac{9.78 - 15.1}{\sqrt{\frac{4.12^2}{9} + \frac{4.28^2}{10}}}$$

$$= \frac{-5.32}{\sqrt{1.88 + 1.83}}$$

$$= \frac{-5.32}{\sqrt{3.71}} = \frac{-5.32}{1.93} \approx -2.76$$

Step 5: Compare the Calculated t-value with the critical t-value

• Degrees of freedom (df): For two sample t-test, we use smaller of $n_1 - 1$ and $n_2 - 1$. Thus $df = 8$

• Critical t-value (from the question) 2.11, for $\alpha = 0.05$ (two-tailed)

Step 6: Decision Rule

If $|t| > 2.11$, reject the null hypothesis

Since the Calculated $t = -2.76$ and $|t| = 2.76 > 2.11$, we reject the

Step-7: Conclusion

There is significant difference between the marks of boys and girls at 0.05 Significance level

Chi-Square test

Dec 2023

- 1) Calculate the chi-square value for the following data of incidences of Water borne diseases in three tropical regions (8 marks)

	India	Equador	South America	Total
Typhoid	31	14	45	90
cholera	2	5	53	60
Diarrrhoea	53	45	2	100
	86	64	100	250

Solⁿ:

Step 1: State the hypothesis

- NULL hypothesis: There is no significant association between (H₀) the tropical regions and the indices of Water borne diseases
- Alternate hypothesis: There is a significant association between the tropical regions and the indices of Water borne diseases.

Step 2: Calculate the Expected Frequencies

The expected frequency for each cell is calculated using formula

$$E = \frac{\text{Row total} \times \text{Column total}}{\text{Grand total}}$$

Let's calculate the expected values for each cell in the table

For typhoid

$$E_{11}(\text{India}) = \frac{90 \times 86}{250}$$

$$= 30.96$$

$$E_{12}(\text{Ecuador}) = \frac{90 \times 64}{250}$$

$$= 23.04$$

$$E_{13}(\text{South America}) = \frac{90 \times 100}{250}$$

$$= 36.00$$

For cholera

$$E_{21}(\text{India}) = \frac{60 \times 86}{250}$$

$$= 20.64$$

$$E_{22}(\text{Ecuador}) = \frac{60 \times 64}{250}$$

$$= 15.36$$

$$E_{23}(\text{South America}) = \frac{60 \times 100}{250}$$

$$= 24.00$$

For diarrhoea

$$E_{31}(\text{India}) = \frac{100 \times 86}{250}$$

$$= 34.40$$

$$E_{32}(\text{Ecuador}) = \frac{100 \times 64}{250}$$

$$= 25.60$$

$$E_{33}(\text{South America}) = \frac{100 \times 100}{250}$$

$$= 40.40$$

Step 3: Compare Chi-Square Statistics

The chi-square formula is

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

where O is observed value and E is the Expected value

For typhoid

$$\frac{(31 - 30.96)^2}{30.96} = \frac{0.0016}{30.96} = 0.000052$$

$$\frac{(14 - 23.04)^2}{23.04} = \frac{81.5376}{23.04} = 3.54$$

$$\frac{(45 - 36.00)^2}{36.00} = \frac{81.00}{36.00} = 2.25$$

For cholera

$$\frac{(2 - 20.64)^2}{20.64} = \frac{352.9216}{20.64} = 17.10$$

$$\frac{(5 - 15.36)^2}{15.36} = \frac{107.7696}{15.36} = 7.02$$

$$\frac{(53 - 24.00)^2}{24.00} = \frac{841.00}{24.00} = 35.04$$

For diarrhoea

$$\frac{(53 - 34.40)^2}{34.40} = \frac{345.96}{34.40} = 10.06$$

$$\frac{(45 - 25.60)^2}{25.60} = \frac{380.25}{25.60} = 14.85$$

Step 4: Sum the Chi-Square values

$$\chi^2 = 0.000052 + 3.54 + 2.25 + 17.10 + 7.02 + 35.04 + 10.06 + 14.85 + 36.10$$
$$= 125.96$$

Step 5: Determine the Degrees of freedom

The degrees of freedom (df) are calculated as

$$df = (\text{number of rows} - 1) \times (\text{number of columns} - 1)$$

In these Case there are 3 rows and 3 column so

$$df = (3 - 1) \times (3 - 1)$$

$$= 2 \times 2$$

$$= 4$$

Step 6: Compare the Chi-Square value with Critical value

To determine, if the result is significant, we compare the calculated Chi-square value with the critical value from the chi-square distribution table at a significance level of 0.05 for 4 degrees of freedom.

The critical value is approximately 9.488

Since the Calculated value i.e. χ^2 value = 125.96 is much greater than critical value we reject the null hypothesis

$\therefore$ there is significant association between the tropical regions and incidence of Water borne diseases

June 2023

2) Calculate χ^2 value using the table below. Critical $\chi^2 = 7.851$ (7 marks)

	Observed Frequency		Expected frequency	
	Female	Male	Female	Male
Without graduation	6	7	6.08	6.92
College	13	16	13.55	15.45
Bachelors degree	16	15	14.49	16.51
Master's degree	8	11	8.88	10.12
Total	43	49	43	49

Soln:

Step 1: State the hypothesis

• NULL Hypothesis (H_0): There is no significant difference between the observed and Expected frequencies

• Alternate Hypothesis (H_1): There is a significant difference between the observed and Expected frequencies

Step 2: Chi-Square formula

Chi-Square Statistics is Calculated using formula

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where

O_i = observed frequency

E_i = Expected frequency

Step 3: Compute chi-Square for each pair

we will now calculate $\frac{(O_i - E_i)^2}{E_i}$ for each observed and expected frequency

1. Without Graduation

a. Female $\frac{(6 - 6.08)^2}{6.08} = \frac{(-0.08)^2}{6.08} = 0.00105$

2. College

$$\bullet \text{ Female } \frac{(13 - 13.55)^2}{13.55} = \frac{(-0.55)^2}{13.55} = \frac{0.3025}{13.55} = 0.02233$$

$$\bullet \text{ Male } \frac{(16 - 15.45)^2}{15.45} = \frac{(0.55)^2}{15.45} = \frac{0.3025}{15.45} = 0.01959$$

3. Bachelor's degree

$$\bullet \text{ Female } \frac{(16 - 14.49)^2}{14.49} = \frac{(1.51)^2}{14.49} = \frac{2.2801}{14.49} = 0.15737$$

$$\bullet \text{ Male } \frac{(15 - 16.51)^2}{16.51} = \frac{(-1.51)^2}{16.51} = \frac{2.2801}{16.51} = 0.13807$$

4. Master's degree

$$\bullet \text{ Female } \frac{(8 - 8.88)^2}{8.88} = \frac{(-0.88)^2}{8.88} = \frac{0.7744}{8.88} = 0.08723$$

$$\bullet \text{ Male } \frac{(11 - 10.12)^2}{10.12} = \frac{(0.88)^2}{10.12} = \frac{0.7744}{10.12} = 0.07652$$

Step 4: Calculate total χ^2 chi-square

Now, sum up all the calculated values to get the total chi-square value

$$\begin{aligned}\chi^2 &= 0.00105 + 0.00092 + 0.02233 + 0.01959 + 0.15737 + 0.13807 \\ &\quad + 0.08723 + 0.07652 \\ &= 0.50272\end{aligned}$$

Step 5: Compare with Critical Chi-Square value

Critical chi-square value given is 7.851. Since the Calculated chi-square value (0.50272) is less than critical value. We fail to reject null hypothesis. There means there is no significant difference

June 2023

- 8) From the given data in the following table, find out whether there is any relationship between gender and ^{preference}~~performance~~ of the color. Critical value ($\chi^2_{0.05} = 5.991$)

(6 marks)

Color	Male	Female
Red	25	45
Blue	45	25
Green	50	10
Total	120	80

Soln:

Step 1: State the hypothesis

- Null hypothesis (H_0): There is no significant relationship between gender and color preference
- Alternate hypothesis (H_1): There is a significant relationship between gender and color preference

Step 2: Calculated Expected frequencies

Expected frequency for each cell is calculated as

$$E_{ij} = \frac{(\text{row total}) \times (\text{column total})}{\text{Grand total}}$$

$$\begin{aligned}\text{Grand total} &= 120 + 80 \\ &= 200\end{aligned}$$

Color	Male	Female	Column Total
Red	25	45	70
Blue	45	25	70
Green	50	10	60
Row total	120	80	

1. Red

• Expected frequency for male: $\frac{(120 \times 70)}{200} = \frac{8400}{200} = 42$

• Expected frequency for female: $\frac{(80 \times 70)}{200} = \frac{5600}{200} = 28$

2. Blue

• Expected frequency for male: $\frac{(120 \times 70)}{200} = 42$

• Expected frequency for female: $\frac{(80 \times 70)}{200} = 28$

3. Green

• Expected frequency for male: $\frac{120 \times 60}{200} = 36$

• Expected frequency for female: $\frac{80 \times 60}{200} = 24$

Step 3: Compute the chi-square for each pair

we now calculate $\frac{(O_i - E_i)^2}{E_i}$ for each observed and Expected frequency

1. Red

• Male: $\frac{(25 - 42)^2}{42} = \frac{(-17)^2}{42} = \frac{289}{42} = 6.88$

• Female: $\frac{(45 - 28)^2}{28} = \frac{(17)^2}{28} = \frac{289}{28} = 10.32$

2. Blue

• Male: $\frac{(45 - 42)^2}{42} = \frac{(3)^2}{42} = \frac{9}{42} = 0.21$

Green

$$\bullet \text{ Male : } \frac{(50-36)^2}{36} = \frac{(14)^2}{36} = \frac{196}{36} = 5.44$$

$$\bullet \text{ Female : } \frac{(10-24)^2}{24} = \frac{(-14)^2}{24} = \frac{196}{24} = 8.17$$

Step 4: Chi-square test formula

The chi-square statistics is calculated using the formula

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where O_i = observed frequency

E_i = Expected frequency

Now, sum all the calculated values to get the total chi-square value

$$\chi^2 = 6.88 + 10.32 + 0.21 + 0.32 + 5.44 + 8.17$$

$$= 31.34$$

Step 5: Compare with the critical chi-square value

The critical chi-square value given is 5.991. Since the calculated chi-square value (31.34) is greater than critical value, we reject the null hypothesis. This means there is a significant relationship between gender and color preference.

Final Answer

$$\chi^2 = 31.34$$

There is a relationship between gender and color preference

- 4) In an anti-malaria Campaign in a certain area quinine was administered to 812 people out of a total 3248. The number of fever cases is shown below. $\chi^2_{0.05} = 3.84$

Treatment	Fever (A)	No Fever (a)	Total
Quinine (B)	140	30	170
No Quinine (b)	60	20	80
Total	200	50	250

Solⁿ: It is asked to perform a chi-square test to determine if there is significant relationship between the treatment (quinine vs no quinine) and occurrence of fever. Let's solve this step by step.

Step 1: State the Hypothesis

- Null hypothesis (H_0): There is no significant relationship between the treatment and occurrence of fever
- Alternate hypothesis (H_1): There is a significant relationship between the treatment and occurrence of fever

Step 2: Calculate Expected frequencies

The expected frequency for each cell is

$$E_{ij} = \frac{(\text{row total} \times \text{column total})}{\text{Grand total}}$$

$$\begin{aligned} \text{Grand total} &= 200 + 50 \\ &= 250 \end{aligned}$$

For "Fever" (A):

• Expected frequency for "Quinine" (B) with "Fever" (A):

$$= \frac{170 \times 200}{250} = \frac{34000}{250} = 136$$

• Expected frequency for "No Quinine" (b) with "Fever" (A):

$$= \frac{80 \times 200}{250} = \frac{16000}{250} = 64$$

For "No Fever" (a):

• Expected frequency for "Quinine" (B) with "No Fever" (a):

$$= \frac{170 \times 50}{250} = \frac{8500}{250} = 34$$

• Expected frequency for "No Quinine" (b) with "No Fever" (a):

$$= \frac{80 \times 50}{250} = \frac{4000}{250} = 16$$

Step 3: Compute the Chi-Square for each pair

Now, we Calculate $\frac{(O_i - E_i)^2}{E_i}$ for each observed and Expected frequency

(1) For Fever (A):

• Quinine (B):

$$\frac{(140 - 136)^2}{136} = \frac{(4)^2}{136} = \frac{16}{136} = 0.1176$$

• No Quinine (b):

$$\frac{(60 - 64)^2}{64} = \frac{(-4)^2}{64} = \frac{16}{64} = 0.25$$

(2) For No Fever (A):

• Quinine (B):

Observed	Quinine	No Quinine	Total
140	60	170	
20	50	70	
160	110	240	
Expected	136	34	
64	16	80	
100	50	150	

No Quinine (b):

$$\frac{(20-16)^2}{16} = \frac{(4)^2}{16} = \frac{16}{16} = 1$$

Step 4: Calculate total chi-square

Now, Sum all the calculated values to get total chi-square value

$$\chi^2 = 0.1176 + 0.25 + 0.4706 + 1$$

$$= 1.8382$$

Step 5: Compare with the Critical chi-square value

Critical chi-square value is given as $\chi^2_{0.05} = 3.84$. Since

The Calculated Chi-square value is 1.8382 which is less than the Critical value (3.84), we fail to reject null hypothesis

Final Answer

There is no significant relationship between the treatment (quinine vs no quinine) and occurrences of fever based on the data provided

$$\chi^2 = 1.8382$$

done in class

5. A school principal would like to know which days of the week students are most likely to be absent. The principal expects that students will be absent equally during the 5-days week. The principal selects random sample of 100 teachers asking them which day of the week they had the highest number of student absences. The observed and expected results are shown in the table below. Based on these results, do the days for the highest number of absences occur with equal frequencies (Use 5% Significance level)

	Monday	Tuesday	Wednesday	Thursday	Friday
Observed Absences	23	16	14	19	28
Expected Absences	20	20	20	20	20

Step 1: State the hypothesis

- Null hypothesis (H_0): Students are absent equally during 5-days week
- Alternate hypothesis (H_1): Students are not absent equally during 5-days week

Step 2: Calculate chi-square for each

For each day, Compute $\frac{(O_i - E_i)^2}{E_i}$

Class	Observed frequency	Expected frequency
A	23	20
B	16	20
C	14	20
D	19	20
E	28	20

1. Monday $\frac{(23-20)^2}{20} = \frac{(3)^2}{20} = \frac{9}{20} = 0.45$

2. Tuesday $\frac{(16-20)^2}{20} = \frac{(4)^2}{20} = \frac{16}{20} = 0.8$

3. Wednesday $\frac{(14-20)^2}{20} = \frac{(6)^2}{20} = \frac{36}{20} = 1.8$

4. Thursday $\frac{(19-20)^2}{20} = \frac{(-1)^2}{20} = \frac{1}{20} = 0.05$

5. Friday $\frac{(28-20)^2}{20} = \frac{(8)^2}{20} = \frac{64}{20} = 3.2$

Step 3: Calculate the total chi-square value

Summing all the values:

$$\chi^2 = 0.45 + 0.8 + 1.8 + 0.05 + 3.2$$
$$= 6.3$$

Step 4: Compare with the Critical value

If we are given Critical value then we compare this calculated value with it. Now let's calculate Critical values by using degrees of freedom ($df = n - 1$) $df = 4$ and at 5% significance level

Since $\chi^2 = 6.3$ is less than critical value (9.488), we fail to reject null hypothesis. This means absence is distributed roughly equally across the week.

6. Find the value of the chi-square for the following information:

Class	A	B	C	D	E
Observed frequency	8	29	44	12	5
Expected frequency	23	5	34	21	9

Solⁿ:

Calculate chi-square for each class

• class A $\frac{(8-23)^2}{23} = \frac{(-15)^2}{23} = \frac{225}{23} = 9.78$

• class B $\frac{(29-5)^2}{5} = \frac{(24)^2}{5} = \frac{576}{5} = 115.2$

• class C $\frac{(44-34)^2}{34} = \frac{(10)^2}{34} = \frac{100}{34} = 2.94$

• class D $\frac{(12-21)^2}{21} = \frac{(-9)^2}{21} = \frac{81}{21} = 3.86$

• class E $\frac{(5-9)^2}{9} = \frac{(-4)^2}{9} = \frac{16}{9} = 1.78$

Calculate total chi-square value

$$\chi^2 = 9.78 + 115.2 + 2.94 + 3.86 + 1.78$$

$$= 133.56$$

(Done in class)

1. A milk producers union wishes to test whether the preference pattern of the consumer for its product is dependent on income levels. A random sample of 500 individuals gives the following data:

Income	Product preferred		
	Product A	Product B	Product C
Low	170	30	80
Medium	50	25	60
High	20	10	55

Can you conclude that the preference patterns are independent of income levels. Use the χ^2 value at 5% level of significance

Step 1: Define hypothesis

- NULL hypothesis (H_0): The preference for products is independent of income levels.
- Alternate hypothesis (H_1): The preference for products depends on income levels.

Step 2: Set Significance level

The Significance level (α) is 0.05

Step 3: Observed frequency table

The given table shows the observed frequencies

Income	Product A	Product B	Product C	Column Total
Low	170	30	80	280
Medium	50	25	60	135
High	20	10	55	85
Row Total	240	65	195	500 ← Grand total

Step 4: Calculate Expected frequencies

The expected frequency for each cell is calculated using the following formula:

$$E_{ij} = \frac{(\text{Row total} \times \text{Column total})}{\text{Grand total}}$$

product A	product B	product C
<u>Low Income</u> $E_{11} = \frac{280 \times 240}{500} = 134.4$	<u>Low Income</u> $E_{21} = \frac{280 \times 65}{500} = 36.4$	<u>Low Income</u> $E_{31} = \frac{280 \times 195}{500} = 109.2$
<u>Medium Income</u> $E_{12} = \frac{135 \times 240}{500} = 64.8$	<u>Medium Income</u> $E_{22} = \frac{135 \times 65}{500} = 17.55$	<u>Medium Income</u> $E_{32} = \frac{135 \times 195}{500} = 52.65$
<u>High Income</u> $E_{13} = \frac{85 \times 240}{500} = 40.8$	<u>High Income</u> $E_{23} = \frac{85 \times 65}{500} = 11.05$	<u>High Income</u> $E_{33} = \frac{85 \times 195}{500} = 33.15$

Step 5: Compute chi-Square for each income and product type

product A	product B
<u>Low</u> $\frac{(170 - 134.4)^2}{134.4} = \frac{1267.66}{134.4} = 9.42$	<u>Low</u> $\frac{(30 - 36.4)^2}{36.4} = \frac{40.96}{36.4} = 1.12$
<u>Medium</u> $\frac{(50 - 64.8)^2}{64.8} = \frac{219.04}{64.8} = 3.38$	<u>Medium</u> $\frac{(25 - 17.55)^2}{17.55} = \frac{55.5}{17.55} = 3.16$
<u>High</u> $\frac{(20 - 40.8)^2}{40.8} = \frac{432.64}{40.8} = 10.60$	<u>High</u> $\frac{(10 - 11.05)^2}{11.05} = \frac{1.10}{11.05} = 0.09$

Product C

$$\text{low} = \frac{(80 - 109.2)^2}{109.2} \quad \text{Medium} = \frac{(60 - 52.95)^2}{52.95} \quad \text{High} = \frac{(55 - 33.15)^2}{33.15}$$

$$= \frac{852.64}{109.2}$$

$$= \frac{49.70}{52.95}$$

$$= \frac{477.42}{33.15}$$

$$= 7.80$$

$$= 0.938$$

$$= 14.40$$

Overall Chi-Square value

$$\chi^2 = 46.978$$

Step 6: Degrees of freedom

The degrees of freedom are calculated as

$$df = (r-1) \times (c-1)$$

$$= (3-1) \times (3-1)$$

$$= 2 \times 2$$

$$= 4$$

Step 7: Compare χ^2 with critical value

The critical value for χ^2 with 4 degrees of freedom at 0.05 Significance level is 9.488

Since Calculated χ^2 is much larger than the critical value

There is enough evidence to conclude that the preference for product is depends on income level

Treatment	Favorable response	No response	Total
New treatment	40	50	90
Conventional treatment	60	40	100
Total	100	90	190

8. On the basis of information given below about the treatment of ²¹⁰~~200~~ patients suffering from a disease. State whether the new treatment is comparatively superior to the conventional treatment.

Treatment	No. of patients	
	Favourable response	No response
New treatment	90	20
Conventional treatment	60	40

For drawing your inference, is the value of χ^2 for one degree of freedom at 5% level of significance as 3.84

Soln:

Step 1: Define hypothesis

• Null hypothesis (H_0): The new treatment is not superior to the conventional treatment (both treatments have same response rate)

• Alternate hypothesis (H_1): The new treatment is superior to the conventional treatment (higher favourable response rate)

Step 2: Set the Significance level

Significance level (α) is 0.05

Step 3: Create observed frequency table

Treatment	Favourable response	No response	total
New treatment	90	20	110
Conventional treatment	60	40	100
Total	150	60	210

Step 4: Calculate expected frequencies

Expected frequency for each cell is calculated using the formula

$$E_{ij} = \frac{(\text{Row total} \times \text{column total})}{\text{Grand total}}$$

Row 1 (New treatment)

• For favourable response

$$E_{11} = \frac{110 \times 150}{210} \quad E_{11} = \frac{110 \times 150}{200} \\ = 78.5$$

• For no response

$$E_{12} = \frac{110 \times 60}{200} \\ = 33 \\ 31.42$$

Row 2 (Conventional Treatment)

• For favourable response

$$E_{21} = \frac{100 \times 150}{200} \\ = 75 \\ 71.4$$

• For no response

$$E_{22} = \frac{100 \times 60}{200} \\ = 30 \\ 28.57$$

Step 5: Compute the Chi-Square Statistics

Using the formula for χ^2

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

O_{ij} = observed frequency

E_{ij} = Expected frequency

Row 1 (New treatment)

• For favourable response

$$\frac{(90 - 78.5)^2}{78.5} \\ = 1.68$$

• No response

$$\frac{(20 - 31.42)^2}{31.42}$$

Row 2 (Conventional treatment)

• For favourable response

$$\frac{(60 - 75)^2}{75} + \frac{(60 - 71.4)^2}{71.4} \\ = 1.82$$

• For No response

$$\frac{(40 - 28.57)^2}{28.57}$$

Overall chi-square value

$$\chi^2 = 0.6818 + 5.12 + 3 + 9$$

$$= 17.80$$

$$12.22$$

Step 6: Degrees of freedom

$$df = (r-1)(c-1)$$

$$= (2-1)(2-1)$$

$$= 1$$

Step 7: Compare χ^2 with Critical value

Critical value for χ^2 with 1 degree of freedom at a 0.05

Significance level is 3.84

Since the Calculated χ^2 value (~~17.80~~^{12.22}) is larger than critical value, we reject null hypothesis

Conclusion

There is enough evidence to conclude that the new treatment is superior to the conventional treatment with a higher favorable response rate

Numerical on the Concept of ANOVA

- 1) To assess the Significance of possible variation in a certain test between Convent schools of city, a Common test was given to the number of Students taken at a random from the fifth class of 3 schools concerned, the results are given below $F_{0.05} = 3.89$

(8 marks)

A	B	C
9	13	14
11	12	13
13	10	17
9	15	7
8	5	9

Make Analysis of Variance for the given data

Step 1: Define the hypothesis

Null hypothesis (H_0): There is no significant difference in the mean of test score between the three schools

Alternate hypothesis (H_1): Atleast one school's mean score is significantly different from the others

Step 2: Calculate the mean for each group

• Mean of A ($\bar{X}_1$):

$$\bar{X}_1 = \frac{9+11+13+9+8}{5} = \bar{X}$$

$$\bar{X}_1 = 10$$

1) To assess the significance of possible variation between Convent schools of city, a common test was given to the number of student of 5th class of 3 schools. The results are given below. Test = 3.84

$$\bar{X}_2 = \frac{13+12+10+15+5}{5}$$

$$= \frac{55}{5}$$

(Example)

A	B	C
P	13	11
L	12	13
13	10	14
P	12	17
8	2	9

• Mean of C ($\bar{X}_3$):

$$\bar{X}_3 = \frac{14+13+17+7+9}{5}$$

$$= \frac{60}{5}$$

$$= 12$$

Step 1: Define the hypothesis
Null hypothesis (H_0): There is no significant difference in the

Step 3: Calculate overall Mean (Grand mean)

$$\bar{X} = \frac{(9+11+13+9+8+13+12+10+15+5+14+13+17+7+9)}{15}$$

$$= \frac{165}{15}$$

$$= 11$$

Step 4: Calculate Sum of Squares between groups (SSB)

$$SSB = n_1(\bar{X}_1 - \bar{X})^2 + n_2(\bar{X}_2 - \bar{X})^2 + n_3(\bar{X}_3 - \bar{X})^2$$

$n_1 = n_2 = n_3 = 5$ (number of observations in each group)

$$\bar{X}_1 = 10, \bar{X}_2 = 11, \bar{X}_3 = 12 \text{ and } \bar{X} = 11$$

$$SSB = 5(10-11)^2 + 5(11-11)^2 + 5(12-11)^2$$

$$SSB = 5(1)^2 + 5(0) + 5(1)$$

$$SSB = 5 + 5$$

$$SSB = 10$$

Step 5: Calculate the Sum of Squares within Groups (SSW)

The formula for SSW is

$$SSW = \sum (x_{ij} - \bar{x}_i)^2$$

where x_{ij} are individual observations, and $\bar{x}_i$ is the mean of group

For group A

$$SSW_A = (9-10)^2 + (11-10)^2 + (13-10)^2 + (9-10)^2 + (8-10)^2$$

$$SSW_A = 1 + 1 + 9 + 1 + 4$$

$$SSW_A = 16$$

For group B

$$SSW_B = (13-11)^2 + (12-11)^2 + (10-11)^2 + (15-11)^2 + (5-11)^2$$

$$SSW_B = 4 + 1 + 1 + 16 + 36$$

$$= 58$$

For group C

$$SSW_C = (14-12)^2 + (13-12)^2 + (17-12)^2 + (7-12)^2 + (9-12)^2$$

$$SSW_C = 4 + 1 + 25 + 25 + 9$$

$$= 64$$

$$\begin{aligned}\text{Total } SSW &= SSW_A + SSW_B + SSW_C \\ &= 16 + 58 + 64 \\ &= 138\end{aligned}$$

Step 6: Calculate the degrees of freedom

• Degrees of freedom between groups (df_1)

$$\begin{aligned}k-1 &= 3-1 \\ &= 2\end{aligned}$$

• Degrees of freedom within groups (df_2)

$$\begin{aligned}N-k &= 15-3 \\ &= 12\end{aligned}$$

where $k=3$ (Number of groups)

$N=15$ (Total number of observations)

Step 7: Calculate mean Square between Groups (MSB) and mean Square within groups (MSW)

• mean Square between groups

$$MSB = \frac{SSB}{df_1}$$

$$= \frac{10}{2}$$

$$= 5$$

• Mean Square within groups

$$MSW = \frac{SSW}{df_2}$$

Step 8: Calculate F-statistics

$$F = \frac{MSB}{MSW}$$

$$= \frac{5}{11.5}$$

$$= 0.435$$

Step 9: Compare F-statistics with critical value

The critical value of $F_{0.05}$ is given as 3.89

Since the Calculated F-value is (0.435) is less than Critical value (3.89) we fail to reject null hypothesis

Conclusion

There is no significant difference in the test scores between the three schools at the 5% significance level